

Razia Sultana Jui

Research Interests: Security & Privacy | Human-Centered Computing
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EDUCATION

BRAC University

Bachelor of Science in Computer Science and Engineering
Department of Computer Science and Engineering, Dhaka, Bangladesh

Summer 2018 – Fall 2024

CGPA (Majors): 3.25/4.00

RESEARCH PROJECT EXPERIENCE

HarmShield: Identifying and Mitigating Harms in LLM and VLM-Based Assisting Systems

LLM and VLM Safety, Trauma-Informed Design, Vulnerable system

2026 – Present

- Investigating how widely deployed LLM-based agents respond to user disclosures of trauma, partner abuse, and accident crises, with a focus on failure modes that risk **re-traumatization, unsafe advice, or mishandled escalation**.
- Designing a **structured framework** grounded in safety, trustworthiness, choice, collaboration, and empowerment principles and translating these into measurable criteria for AI system behavior, which assists the victims in the proper way.

Empirical Study of Multimodal Hate Meme Detection: Can VLMs See What Text-Only Models Miss?

Multimodal ML, VLMs, Content Moderation, Online Harms

2025 – Present

- Curated a **117-meme benchmark** of *visual memes* content whose text is superficially benign yet hateful when image and text are interpreted together sourced from Facebook Hateful Memes and M3 datasets where established text-only baselines (EasyOCR + HateBERT, Detoxify, HateXplain) confirming near-zero detection (HateXplain: 0%, Detoxify: 12.9%)
- Benchmarked **14+ open-source VLMs** across five model families (Gemma3, Qwen-VL, LLaVA, InternVL3, SmolVLM, Phi-3.5 Vision) and conducted a **systematic definition-free vs. hate-definition-conditioned study** revealing that prompt framing causes F1 increase up to **+0.637** (LLaVA-1.5-13B: 0.202 $\rightarrow$ 0.839) and best overall: Gemma3-27B at F1 = **0.898**.

Identifying Hate Speech in Bangla Language Text Using Natural Language Processing

| [Link](#)

Undergraduate Thesis – NLP, Deep Learning, Hate Speech Detection, Bangla Text Classification **2024 – 2025**

- Completed undergraduate thesis under the supervision of **Mr. Annajiat Alim Rasel**, Senior Lecturer, Department of Computer Science and Engineering, BRAC University, and developed and compared deep learning models leveraging **BiLSTM, CNN-LSTM**, and transformer-based architectures (**mBERT, XLM-R**) for hate speech detection in Bangla, a low-resource language where automated moderation infrastructure is critically underdeveloped.
- Designed a preprocessing and normalization pipeline for Bangla text (script normalization, code-mixing handling, noisy social-media cleanup) that materially improved classification performance and robustness across domains. which is motivated by the disproportionate online harm faced by women, religious minorities, and dissenting voices in Bangla-speaking online spaces, which aims to support safer digital communication for at-risk users.

PROFESSIONAL PROJECTS

Trenzo E-commerce Website Development | *Next.js, React, Node.js*

2022 | [Website](#)

- Developed a full-stack e-commerce web application for a premium wallet retail brand.
- Implemented product catalog filtering, cart checkout, and an order tracking module.

ACADEMIC PROJECTS

Fire and Smoke Detection System using Arduino | C++, Arduino Uno, DHT22, MQ2

Summer 2022

- Designed an automated safety system to detect fire, smoke, and gas leaks using flame and MQ2 gas sensors to mitigate industrial risks.
- Integrated DHT22 sensors for real-time temperature and humidity monitoring, displaying live data on an LCD interface and Implemented an immediate alert mechanism using buzzers and LED indicators to signal hazard detection effectively.

Heart Disease Prediction using Machine Learning | Python, Scikit-learn, Pandas, NumPy

Fall 2022

- Developed a diagnostic model on a Kaggle dataset of 1,026 entries, implementing KNN, Decision Tree, and Logistic Regression algorithms.
- Achieved 93% accuracy by applying MinMax and Standard Scaler normalization techniques to optimize model performance and evaluated classification reliability using confusion matrices, precision, recall, and F1-scores for robust disease detection.

Real Estate Management System | Django, Python, HTML, CSS, JavaScript

Spring 2021

- Built a full-stack web platform for property trading, featuring a dynamic 48-hour bidding system to optimize sales value.
- Engineered role-based access, enabling secure property listings, price filtering, and user verification, which eliminates intermediaries, reducing transaction costs by 9% for rural property buyers.

HIGHLIGHTED ACADEMIC COURSES

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|------------------------------------|---------------------------------------|
| • CSE-423 Computer Graphics | • CSE-420 Compiler Design |
| • CSE-425 Neural Networks | • CSE-360 Computer Interfacing |

TECHNICAL SKILLS

Languages: Python, Java, C, C++, MATLAB, JavaFX, Bash, Assembly, Swing

ML / NLP: PyTorch, scikit-learn, pandas, NumPy, BERT, RAG, LLM, VLM

Databases: Oracle, MySQL, SQLite, MongoDB

OS: Linux (Ubuntu, Kali), Windows

Version Control: Git, GitHub

Frameworks: Django, Node.js, React, JavaFX

Web: HTML, CSS, PHP, Bootstrap, WordPress, JSON, XML

Compiler Tools: Flex, Bison, EMU 8086

Microcontrollers: Arduino, ATmega 32, Atmel Studio, Proteus 832

Writing: L^AT_EX, Beamer, Overleaf

Other: NS3, iGraphics, Cisco Packet Tracer, OpenGL

REFERENCES

Mr. Annajiat Alim Rasel

Senior Lecturer, Department of Computer Science and Engineering

BRAC University, Dhaka, Bangladesh

Email: annajiat@bracu.ac.bd

Relation: Undergraduate Thesis Supervisor

Dr. Muhammad Iqbal Hossain

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BRAC University, Dhaka, Bangladesh

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Relation: Course Instructor & Research Mentor